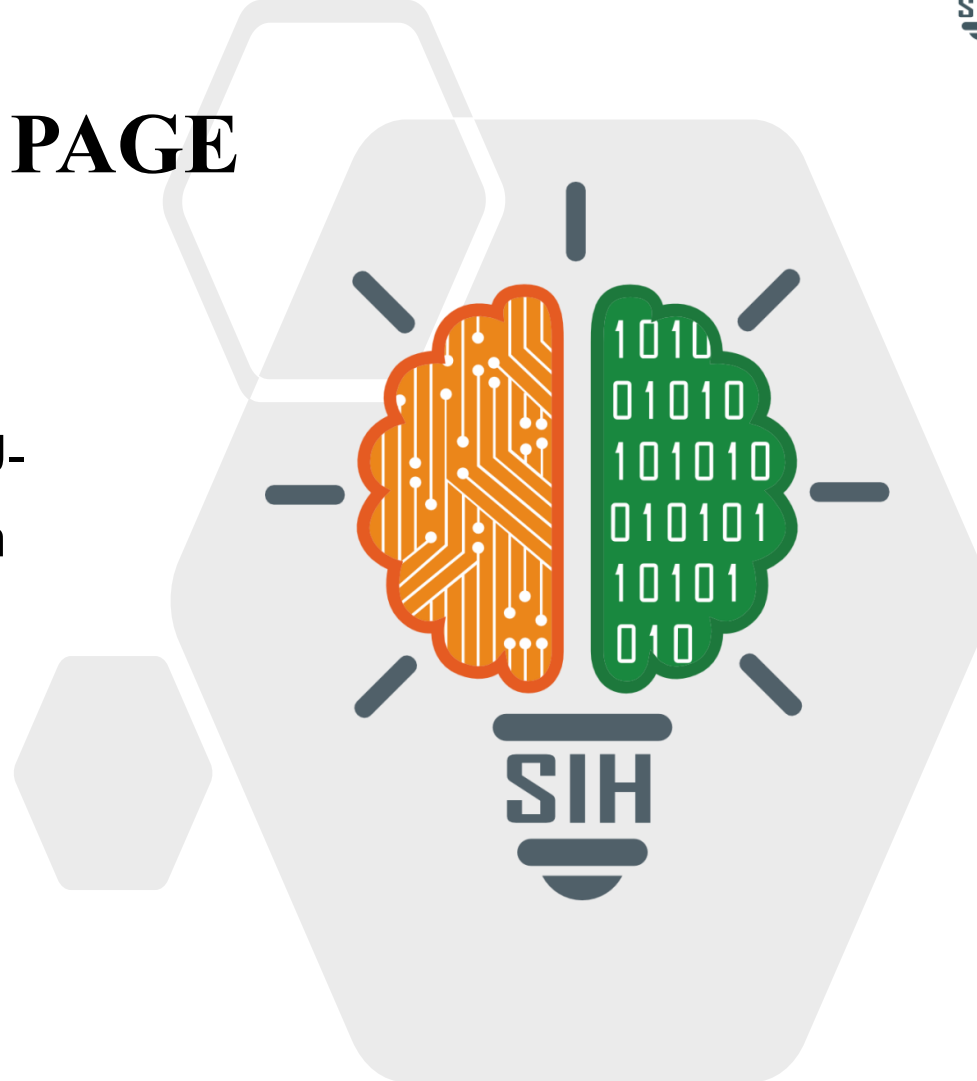


TITLE PAGE

- **Problem Statement ID** – SIH26119
- **Problem Statement Title**- Indigenous GPU-Accelerated Optimization Solver (Sovereign Alternative to Express / CPLEX)
- **Theme**- Smart Automation
- **PS Category**- Software
- **Team ID**- 127364
- **Team Name (Registered on portal)**- VIGHNAX





SAMADHAN: a sovereign optimization solver



PROPOSED SOLUTION

One solver for LP, QP and MILP, written from scratch. A GPU engine solves plant-scale planning models; a C++ core handles the integer decisions.

DETAILED EXPLANATION The GPU engine runs the restarted primal–dual hybrid gradient method (PDLP family) for linear and quadratic programs using only sparse matrix–vector products. The C++17 core runs a bounded dual simplex with branch-and-cut for mixed-integer programs.

HOW IT ADDRESSES THE PROBLEM A 1.16-million-variable crude-to-depot refinery plan solves in 64 s on one laptop GPU, 5.1 × faster than HiGHS at the same accuracy. No CPLEX, Gurobi, Xpress or HiGHS code is inside.

INNOVATION AND UNIQUENESS GPU-first and factorisation-free, so it keeps scaling where simplex and interior point slow down. Every line is open and auditable: a solver core MRPL can inspect, tune and certify.



A recorded run on an RTX 5050 laptop GPU: 1,157,120 variables, certified optimal in 63.9 s.

5.1×

faster than HiGHS,
same accuracy

22×

faster in
fast mode

26

MIPLIB 3 models
proved optimal



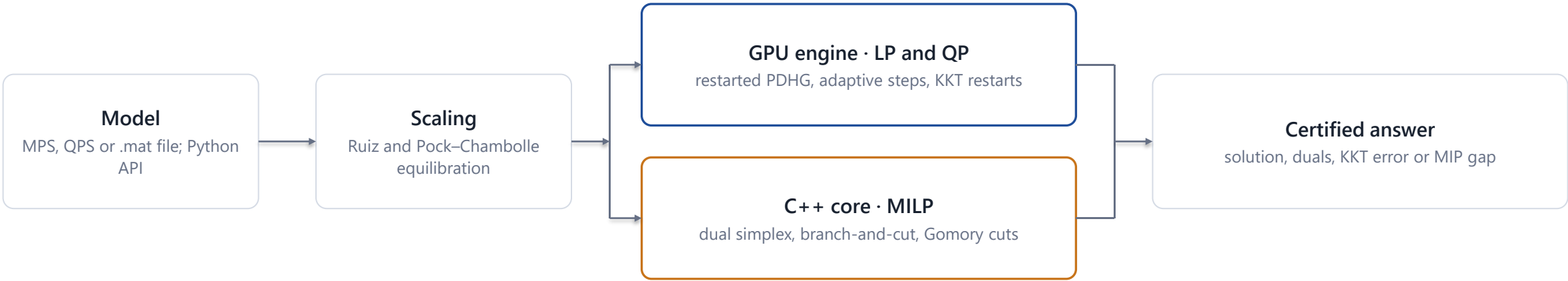
TECHNICAL APPROACH



TECHNOLOGIES

GPU engine · LP and QP	Python + PyTorch CUDA: sparse CSR mat-vecs, 64-iteration CUDA graphs	NVIDIA GPU; same code on CPU
C++ core · MILP	C++17, compiled with zig/clang, C ABI via ctypes	any x86-64 CPU, Windows or Linux
Interfaces	Python API, command line, MPS / QPS / .mat readers	tested on every push (CI)

ARCHITECTURE



METHODOLOGY

- 01

Read and scale
own MPS / QPS reader
- 02

Solve
GPU engine or C++ core
- 03

Certify
KKT error or proven gap
- 04

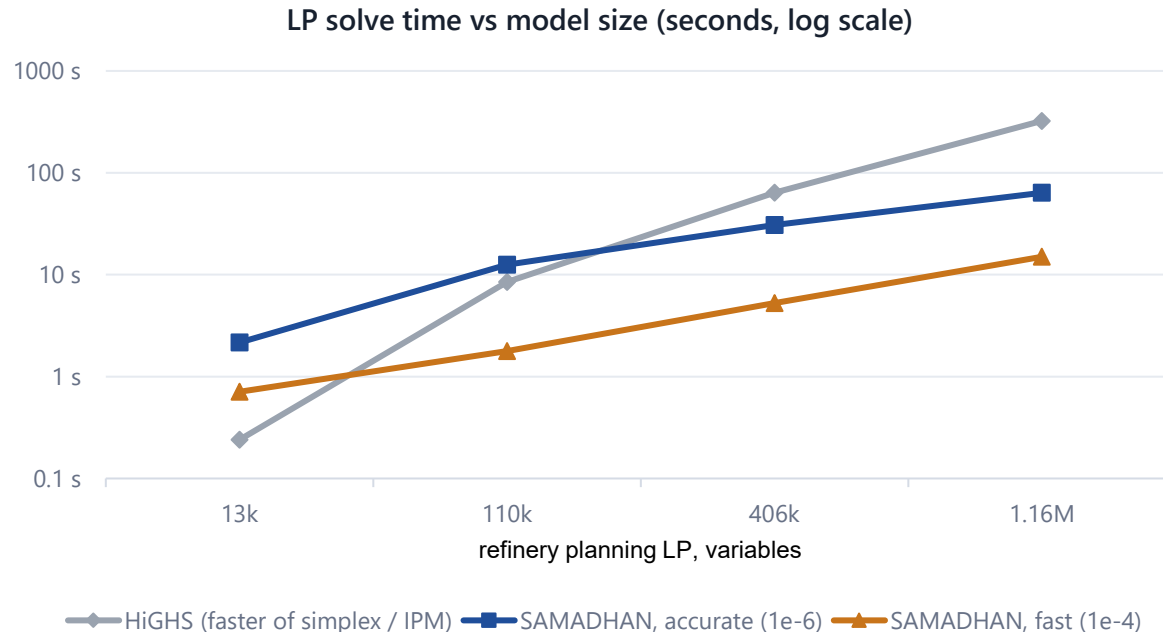
Benchmark
HiGHS on Netlib, MIPLIB, Maros–Mészáros



FEASIBILITY AND VIABILITY



MEASURED ON ONE LAPTOP



STANDARD BENCHMARKS THE PS NAMES

86/91

Netlib LPs

solved to 1e-4 relative KKT

26/58

MIPLIB 3 MILPs

proved optimal; 3 that HiGHS could not finish in 60 s

79/129

Maros–Mészáros QPs

solved; all 64 solved by both agree with HiGHS

78×

refinery QP

faster than HiGHS at 13k variables; HiGHS does not finish from 110k

RISKS AND HOW WE HANDLE THEM

First-order methods stop at moderate accuracy

Stop only on a certified KKT error; crossover to exact vertices in the build phase.

HiGHS still proves 20 more MIPLIB models

Presolve, MIR and cover cuts, and a sparse LU next; hard MILPs are the long-term track.

No GPU at a site

The same code runs on the CPU, and the C++ core needs no GPU at all.



IMPACT AND BENEFITS



POTENTIAL IMPACT ON THE TARGET AUDIENCE

MRPL planning teams

crude selection, blending and product dispatch, re-planned in seconds

Other refineries and petrochemical PSUs

the same planning models, with no licence server

Power, logistics, steel and railways

unit commitment, routing and production scheduling

Strategic and air-gapped sites

runs on-premise; no data leaves the plant

BENEFITS

Economic

No per-core or per-user licence fees;
runs on a commodity GPU or CPU.

Sovereign

Indian-owned code under Apache-2.0
that can be audited, changed and
certified.

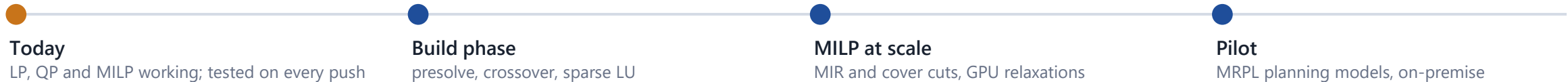
Operational

Plans that took minutes take seconds,
so planners try more what-if scenarios.

Environmental

Better crude and energy plans mean
less fuel use, flaring and waste.

ROADMAP



LP ON GPUS

- 1 Applegate et al. Practical large-scale linear programming using primal-dual hybrid gradient. NeurIPS 2021.
- 2 Lu & Yang. cuPDLP.jl: a GPU implementation of restarted PDHG for linear programming. arXiv:2311.12180, 2023.
- 3 Chambolle & Pock. A first-order primal-dual algorithm for convex problems. J. Math. Imaging Vision 40, 2011.
- 4 Ruiz. A scaling algorithm to equilibrate both rows and columns norms in matrices. RAL-TR-2001-034.

QP AND MILP

- 5 Lu & Yang. A practical and optimal first-order method for large-scale convex QP (PDQP). arXiv:2311.07710.
- 6 Huangfu & Hall. Parallelizing the dual revised simplex method. Math. Prog. Comp. 10, 2018.
- 7 Achterberg. Constraint Integer Programming. PhD thesis, TU Berlin, 2007.
- 8 Balas, Ceria, Cornuéjols, Natraj. Gomory cuts revisited. Oper. Res. Letters 19, 1996.

BENCHMARKS

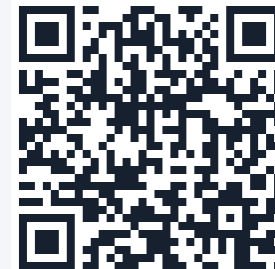
- 9 Bixby, Ceria, McZeal, Savelsbergh. An updated mixed integer programming library: MIPLIB 3.0. Optima 58, 1998.
- 10 Maros & Mészáros. A repository of convex quadratic programming problems. Optim. Methods Softw. 11, 1999.
- 11 Netlib LP collection (netlib.org/lp) and H. Mittelmann's benchmarks (plato.asu.edu/bench.html).



Demo video

Scan the code

3 minutes · youtu.be/zagX0zqtU



Source, benchmarks, tests

Scan the code

[github.com/NAMAN-THAKUR-1944/
SIH2026-SIH26119-SAMADHAN](https://github.com/NAMAN-THAKUR-1944/SIH2026-SIH26119-SAMADHAN)